

Business Plan

Marketing Campaign Decision Tree Classification Assignment

Business Name: AdVantage Growth Studio

Purpose: Use this business plan with the provided Excel dataset to build and interpret a Decision Tree classification model.

1. Business Overview

Item	Description
Business Type	Data-driven campaign advertising and marketing agency
Main Service	Helping small and medium-sized businesses design, target, and evaluate marketing campaigns
Main Customers	Retailers, online stores, local service providers, and small consumer brands
Main Business Goal	Improve campaign conversion rates while reducing wasted advertising budget
Machine Learning Goal	Predict whether a customer is likely to convert after receiving a marketing campaign

AdVantage Growth Studio is a data-driven marketing agency that supports businesses in planning, running, and improving digital marketing campaigns. The agency works with customer data, campaign engagement data, and sales outcomes to help clients understand which customers are most likely to respond positively to a campaign.

2. Business Problem

The key business problem is campaign inefficiency. Many businesses spend money sending campaigns to customers who are unlikely to respond. At the same time, they may fail to identify customers who are ready to buy but need the right message, offer, or channel. This creates two problems: wasted marketing budget and missed revenue opportunities.

- Some customers receive offers but do not convert, which wastes advertising money.
- Some customers would convert if the business targeted them more effectively.
- Marketing teams need a simple, interpretable model that explains why a customer may or may not convert.
- Managers need business-friendly insights, not only technical model results.

3. Machine Learning Decision

Machine learning can help the business decide which customers should receive follow-up marketing attention. For example, the business may use the model to prioritize customers for a discount reminder, a sales follow-up, a personalized email, or a retargeting advertisement.

Recommended use of the model

- Identify customers who are more likely to convert after a campaign.
- Support campaign budget allocation across channels such as Email, Social Media, Search Ads, Display Ads, and SMS.
- Understand which customer behaviors and campaign features are most strongly related to conversion.

- Give managers an interpretable decision-support tool using a Decision Tree model.

Important note: The model should support human decision-making. It should not automatically replace marketing managers, customer service judgment, or ethical review.

4. Dataset Description

The Excel dataset contains synthetic customer-level marketing campaign data. Each row represents one customer who received a marketing campaign. The target variable is whether the customer converted after the campaign.

Dataset Element	Explanation
Observation Unit	One customer campaign record
Number of Rows	605 rows, including 5 intentional duplicate rows for cleaning practice
Target Variable	Converted
Target Meaning	Yes = customer converted after the campaign; No = customer did not convert
Main Feature Types	Customer demographics, customer behavior, campaign channel, campaign type, engagement indicators, loyalty and satisfaction measures
Cleaning Practice	The dataset includes a small number of missing values and duplicate rows so students can practice data preparation

Target variable

Converted is the classification target. The Decision Tree should learn patterns that help predict whether Converted is Yes or No.

Input features

Students may use the customer and campaign variables as input features. However, Customer_ID should not be used as a predictive feature because it is only an identifier and does not represent a meaningful customer characteristic.

- Customer demographics: Age, Annual_Income, Region
- Customer relationship: Customer_Segment, Loyalty_Score, Customer_Satisfaction
- Campaign information: Campaign_Channel, Campaign_Type, Discount_Offered_Percent, Ad_Exposure_Count
- Engagement behavior: Email_Opened, Ad_Clicked, Website_Visits_Last_30_Days, Social_Media_Engagement_Score
- Purchase history: Previous_Purchases, Average_Order_Value, Days_Since_Last_Purchase, Prior_Campaign_Response

5. Why This Prediction Is Useful

Predicting conversion is useful because it turns campaign data into action. A marketing manager can use the prediction to decide where to spend money, which customers need follow-up, and which campaign features may be most effective.

- Better targeting: focus on customers with a higher probability of conversion.
- Budget control: reduce spending on customers who are unlikely to convert.
- Campaign improvement: learn which channels, offers, and engagement behaviors are linked to better results.
- Customer experience: avoid over-contacting customers who are unlikely to respond.

- Business communication: explain results to non-technical managers using feature importance and tree rules.

6. Evaluation Metrics for the Business

For this business problem, F1-score is a strong primary metric because the business wants to balance two risks: **wasting budget on the wrong customers and missing customers who may convert**. Accuracy is useful, but it may not be enough if the classes are imbalanced or if one type of error is more expensive than the other.

Metric	What It Means	Business Interpretation
Accuracy	Overall percentage of correct predictions	Good for a general overview, but may hide important errors.
Precision	Of customers predicted to convert, how many actually converted	Important when the business wants to avoid wasting campaign budget.
Recall	Of customers who actually converted, how many the model found	Important when the business wants to avoid missing potential buyers.
F1-score	Balance between precision and recall	Recommended primary metric when both wasted budget and missed revenue matter.

7. Meaning of False Positives and False Negatives

Error Type	Technical Meaning	Business Meaning
False Positive	The model predicts Converted = Yes, but the customer actually does not convert.	The business may spend money on follow-up marketing for a customer who does not buy. This can waste advertising budget and staff time.
False Negative	The model predicts Converted = No, but the customer actually converts.	The business may miss a valuable customer who could have been targeted more effectively. This can lead to lost revenue opportunities.

8. Expected Important Features

After training the Decision Tree, students should examine feature importance. The most important features may vary depending on the train-test split, preprocessing choices, and tree parameters. However, the following variables are expected to be meaningful from a business perspective:

- Ad_Clicked: clicking an ad is a strong sign of customer interest.
- Website_Visits_Last_30_Days: frequent website visits may indicate purchase intent.
- Loyalty_Score and Previous_Purchases: loyal and repeat customers may be more likely to convert.
- Days_Since_Last_Purchase: customers who purchased recently may behave differently from inactive customers.
- Prior_Campaign_Response: past response can signal future responsiveness.
- Campaign_Channel and Campaign_Type: some channels and campaign designs may perform better than others.
- Discount_Offered_Percent: discounts may increase conversion but can also reduce profit margins.

9. How the Business Can Use the Results

The Decision Tree results should be translated into practical business actions. Students should connect model performance and feature importance to campaign decisions.

Model Insight	Possible Business Action
Ad_Clicked is highly important	Retarget customers who clicked but did not purchase.
Loyalty_Score is highly important	Create special campaigns for loyal customers and retention campaigns for low-loyalty customers.
Campaign_Channel is highly important	Move more budget toward channels with better conversion patterns.
Days_Since_Last_Purchase is highly important	Design reactivation campaigns for customers who have been inactive for a long time.
Discount_Offered_Percent is highly important	Test different discount levels to balance conversion and profitability.

10. Limitations and Possible Bias

This dataset is synthetic and simplified for learning purposes. It does not include every factor that influences real customer behavior. The model may also learn patterns from incomplete or biased data.

- The dataset may not represent all customer groups equally.
- Income and region may introduce fairness concerns if used without careful interpretation.
- The model can identify patterns but does not prove that one feature causes conversion.
- Missing values and duplicate rows may affect the model if they are not handled carefully.
- A Decision Tree can overfit if it becomes too deep and memorizes the training data.

11. Why Human Judgment Is Still Needed

Human judgment is necessary because marketing decisions affect real customers, budgets, and brand reputation. A model can support decisions, but managers must interpret results carefully, check for bias, consider customer experience, and make decisions that align with business strategy and ethical standards.

- Managers should review whether the model recommendations make business sense.
- Humans should check whether the model might disadvantage certain customer groups.
- Marketing teams should consider customer privacy and avoid excessive targeting.
- The final decision should combine model results, business goals, and responsible human oversight.

12. Suggested Student Interpretation Focus

In the assignment notebook, students should not only show code outputs. They should explain the results in business language. A strong submission should connect model metrics, false positives, false negatives, and feature importance to campaign decisions.

- Explain the business problem before building the model.
- Use clear preprocessing steps for categorical and numerical variables.
- Compare training accuracy and testing accuracy to discuss possible overfitting.
- Use the confusion matrix to explain false positives and false negatives.
- Interpret feature importance as business insight, not just a technical output.
- End with one limitation and one responsible AI reflection.